SECTION 1

Installing Node.js

- **LTS baseline:** MCP examples assume Node 20 LTS or newer, which bundles `npm` and `npx` together.
- **Windows options:** Use the official installer from nodejs.org or nvm-windows when multiple Node versions must coexist.
- **macOS path:** Homebrew (`brew install node`) or the macOS installer from nodejs.org are the usual routes.
- **Linux and WSL:** nvm installs user-local Node without sudo; distribution packages via NodeSource are an alternative when nvm is undesirable.

SECTION 2

Verifying Node and npx

- **Version checks:** Fresh shells should report versions for `node`, `npm`, and `npx`, with Node at v20+; “command not found” means PATH or nvm activation was skipped.
- **npx resolution:** `npx` prefers local `node_modules`, then the global npx cache under the user home, downloading from npm only on a cache miss—so the first run is slower than later ones.

SECTION 3

Running an MCP Server with npx

- **Command shape:** Typical launches follow `npx -y <package> [args...]`, where `-y` suppresses the install prompt subprocess clients cannot answer.
- **Filesystem example:** `npx -y @modelcontextprotocol/server-filesystem` plus an allowed directory validates the server in a terminal; it speaks MCP over stdio until stopped with Ctrl-C.
- **Client configuration:** MCP JSON configs set `"command": "npx"` and pass `-y`, the package name, and server-specific arguments as the `args` array—the same pattern works for other npm-hosted MCP servers.

SECTION 4

Troubleshooting and Keeping Node Healthy

- **Common failures:** Missing `npx` after install needs a new shell; EACCES often traces to sudo-installed Node; immediate client exit deserves a direct terminal run to surface stderr; hangs frequently mean `-y` was omitted.
- **Cache hygiene:** `npm cache clean --force` or pinning `@version` on the package name forces a fresh download when a stale cached server misbehaves.
- **Staying current:** Re-run installer, `brew upgrade node`, or `nvm install --lts` periodically, then repeat the three version checks so agents keep using supported Node releases.